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Harmonic and Subharmonic Function Theory on the Hyperbolic Ball

Manfred Stoll



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University of South Carolina



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To Mary Lee

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Preface

The intent of these notes is to provide a detailed and comprehensive treatment of harmonic and subharmonic function theory on hyperbolic space in $\mathbb{R}^n$. Although our primary emphasis will be in the setting of the unit ball $\mathbb{B}$ with hyperbolic metric ds given by

$$ds = \frac{2|dx|}{1 - |x|^2}, \tag{1}$$

we will also consider the analogue of many of the results in the hyperbolic half-space $\mathbb{H}$. Undoubtedly some of the results are known, either in the setting of rank one noncompact symmetric spaces (e.g. [38]), or more generally, in Riemannian spaces (e.g. [13]). An excellent introduction to harmonic function theory on noncompact symmetric spaces can be found in the survey article [47] by A. Koranyi. The 1973 paper by K. Minemura [57] provides an introduction to harmonic function theory on real hyperbolic space considered as a rank one noncompact symmetric space. Other contributions to the subject area in this setting will be indicated in the text.

With the goal of making these notes accessible to a broad audience, our approach does not require any knowledge of Lie groups and only a limited knowledge of differential geometry. The development of the theory is analogous to the approach taken by W. Rudin [72] and by the author [84] in their development of Möbius invariant harmonic function theory on the hermitian ball in $\mathbb{C}^n$. Although our primary emphasis is on harmonic function theory on the ball, we do include many relevant results for the hyperbolic upper half-space $\mathbb{H}$, both in the text and in the exercises. With only one or two exceptions, the notes are self-contained with the only prerequisites being a standard beginning graduate course in real analysis.

In Chapter 1 we provide a brief review of Möbius transformation in $\mathbb{R}^n$. This is followed in Chapter 2 by a characterization of the group $\mathcal{M}(\mathbb{B})$ of

Möbius self-maps of the unit ball $\mathbb{B}$ in $\mathbb{R}^n$. As in [72] we define a family $\{\varphi_a : a \in \mathbb{B}\}$ of Möbius transformations of $\mathbb{B}$ satisfying $\varphi_a(0) = a$, $\varphi_a(a) = 0$, and $\varphi_a(\varphi_a(x)) = x$ for all $x \in \mathbb{B}$. Furthermore, for every $\psi \in \mathcal{M}(\mathbb{B})$, it is proved that there exists $a \in \mathbb{B}$ and an orthogonal transformation A such that $\psi = A\varphi_a$. When $n = 2$, the mappings φ_a correspond to the usual analytic Möbius transformations of the unit disc $\mathbb{D}$ given by

$$\varphi_a(z) = \frac{a - z}{1 - \bar{a}z}. \quad (2)$$

Some of the properties of the mappings $\{\varphi_a\}$ and of functions in $\mathcal{M}(\mathbb{B})$ are developed in Section 2.1. In this chapter we also introduce the hyperbolic metric in $\mathbb{B}$ and in the hyperbolic half-space $\mathbb{H}$. Most of the results of these two sections are contained in the works of L. V. Ahlfors [4], [5], and the text by A. F. Beardon [11].

In Chapter 3 we derive the Laplacian, gradient, and measure on $\mathbb{B}$ that are invariant under $\mathcal{M}(\mathbb{B})$. Even though the formula for the Laplacian can be derived from the hyperbolic metric, we will follow the approach of W. Rudin [72, Chapter 4]. For $f \in C^2(\mathbb{B})$ we define $\Delta_h f$ by

$$\Delta_h f(a) = \Delta(f \circ \varphi_a)(0),$$

where Δ is the usual Laplacian in $\mathbb{R}^n$. The operator Δ_h is shown to satisfy $\Delta_h(f \circ \psi)(x) = (\Delta_h f)(\psi(x))$ for all $\psi \in \mathcal{M}(\mathbb{B})$. Furthermore, an explicit computation gives

$$\Delta_h f(x) = (1 - |x|^2)^2 \Delta f(x) + 2(n-2)(1 - |x|^2) \langle x, \nabla f(x) \rangle,$$

where ∇f is the Euclidean gradient of the function f . In this chapter it is also proved that the Green's function for Δ_h is given by $G_h(x, y) = g(|\varphi_x(y)|)$, where g is the radial function on $\mathbb{B}$ defined by

$$g(r) = \frac{1}{n} \int_r^1 \frac{(1 - s^2)^{n-2}}{s^{n-1}} ds.$$

In Theorem 3.3.1 we prove that for $\psi \in \mathcal{M}(\mathbb{B})$, the Jacobian J_ψ of the mapping ψ satisfies

$$|J_\psi(x)| = \frac{(1 - |\psi(x)|^2)^n}{(1 - |x|^2)^n}.$$

From this it now follows that the Möbius invariant measure τ on $\mathbb{B}$ is given by

$$d\tau(x) = (1 - |x|^2)^{-n} dv(x),$$

where v is the normalized volume measure on $\mathbb{B}$. In the exercises we develop the invariant Laplacian, Green's function, and invariant measure on $\mathbb{H}$.

A real-valued C^2 function f on $\mathbb{B}$ is defined to be either $\mathcal{H}$ -harmonic or $\mathcal{H}$ -subharmonic on $\mathbb{B}$ depending on whether $\Delta_h f = 0$ or $\Delta_h f \geq 0$. It is well known that a continuous function f is harmonic in the unit disc $\mathbb{D}$ if and only if for all r , $0 < r < 1$, and $w \in \mathbb{D}$,

$$f(w) = \frac{1}{2\pi} \int_0^{2\pi} f(\varphi_w(re^{it})) dt, \quad (3)$$

where φ_w is the Möbius transformation of $\mathbb{D}$ given by (2). The above is called the **invariant mean-value property**. One of the first results proved in Chapter 4 is the following analogue of the invariant mean-value property: A real-valued C^2 function f is $\mathcal{H}$ -subharmonic on $\mathbb{B}$ if and only if for all $a \in \mathbb{B}$ and $0 < r < 1$,

$$f(a) \leq \int_{\mathbb{S}} f(\varphi_a(rt)) d\sigma(t), \quad (4)$$

with equality if and only if f is $\mathcal{H}$ -harmonic on $\mathbb{B}$. In the above, $\mathbb{S}$ is the unit sphere in $\mathbb{R}^n$, σ is normalized surface measure on $\mathbb{S}$, and φ_a is the Möbius transformation of $\mathbb{B}$ mapping 0 to a with $\varphi_a(\varphi_a(x)) = x$. The integral in (4) is an average of f over the hyperbolic or non-Euclidean sphere $\{\varphi_a(rt) : t \in \mathbb{S}\}$ whose hyperbolic center is a . Inequality (4) is then used in Section 4.3 to extend the definition of $\mathcal{H}$ -subharmonic to the class of upper semicontinuous functions on $\mathbb{B}$. The remainder of the chapter is devoted to extending some of the standard results about subharmonic functions to $\mathcal{H}$ -subharmonic functions on $\mathbb{B}$. We conclude the chapter with a discussion of quasi-nearly $\mathcal{H}$ -subharmonic functions and prove several inequalities involving these functions that will prove useful later in the text.

The Poisson kernel P_h for Δ_h is introduced in Chapter 5. In Section 5.1 we prove using Green's formula that for $(a, t) \in \mathbb{B} \times \mathbb{S}$,

$$P_h(a, t) = - \lim_{r \rightarrow 1} nr^{n-1}(1-r^2)^{2-n} \langle \nabla G_a(rt), t \rangle,$$

where $G_a(rt) = G_h(a, rt)$ is the Green's function for Δ_h . This immediately gives

$$P_h(x, t) = \left(\frac{1 - |x|^2}{|x - t|^2} \right)^{n-1}, \quad (x, t) \in \mathbb{B} \times \mathbb{S}.$$

The standard results for Poisson integrals of continuous functions are included in Section 5.3, and in Section 5.2 we prove a result of P. Jaming [43] that provides an integral representation of the Euclidean Poisson kernel in terms of the hyperbolic Poisson kernel. In Section 5.5 we investigate the eigenfunctions of Δ_h . We close the section with a brief discussion of the Poisson kernel on $\mathbb{H}$.

In Chapter 6 we consider the spherical harmonic expansions of $\mathcal{H}$ -harmonic functions. One of the key results of this section is that if p_α is a spherical harmonic of degree α on $\mathbb{S}$, then the Poisson integral $P_h[p_\alpha]$ of p_α is given by

$$P_h[p_\alpha](x) = |x|^\alpha S_{n,\alpha}(|x|) p_\alpha\left(\frac{x}{|x|}\right),$$

where $S_{n,\alpha}$ is given by a hypergeometric function. Interestingly, when n is even, $S_{n,\alpha}(r)$ is simply a polynomial in r of degree $n - 2$. These results are then used to show how the Poisson integral $P_h[q]$ can be computed for any polynomial q on $\mathbb{S}$. As an example, in $\mathbb{R}^4$, the $\mathcal{H}$ -harmonic function with boundary values t_1^2 is given by $P_h[t_1^2](x) = \frac{1}{4} + (2 - |x|^2)(x_1^2 - \frac{1}{4}|x|^2)$. In contrast, the Euclidean harmonic function h with boundary values t_1^2 is given by $h(x) = \frac{1}{4}(1 - |x|^2) + x_1^2$. Finally, in Section 6.3 we follow the methods of P. Ahern, J. Bruna, and C. Cascante [2] to derive the spherical harmonic expansion of $\mathcal{H}$ -harmonic functions on $\mathbb{B}$.

Chapter 7 is devoted to the study of Hardy and Hardy–Orlicz type spaces of $\mathcal{H}$ -harmonic and $\mathcal{H}$ -subharmonic functions on $\mathbb{B}$. In Chapter 8, we study the boundary behavior of Poisson integrals on $\mathbb{B}$. This chapter contains many of the standard results concerning non-tangential and radial maximal functions. In addition to proving the usual Fatou theorem (Theorem 8.3.3) concerning non-tangential limits of Poisson integrals of measures, we also include a proof of a local Fatou theorem of I. Privalov [68] for $\mathcal{H}$ -harmonic functions on $\mathbb{B}$.

The Riesz decomposition theorem for $\mathcal{H}$ -subharmonic functions is proved in Chapter 9. The main result of this chapter (Corollary 9.1.3) proves that if f is $\mathcal{H}$ -subharmonic on $\mathbb{B}$ and f has an $\mathcal{H}$ -harmonic majorant, then

$$f(x) = F_f(x) - \int_{\mathbb{B}} G_h(x, y) d\mu_f(y),$$

where μ_f is the Riesz measure of f and F_f is the least $\mathcal{H}$ -harmonic majorant of f . In Section 9.2 we include several applications of the Riesz decomposition theorem, including a Hardy–Stein identity for non-negative $\mathcal{H}$ -subharmonic functions for which f^p , $p \geq 1$, has an $\mathcal{H}$ -harmonic majorant on $\mathbb{B}$. In Section 9.3 we extend a result of D. H. Armitage [8] concerning the integrability of non-negative superharmonic functions. We conclude the chapter by proving that invariant Green potentials of measures have radial limit zero almost everywhere on $\mathbb{S}$, and provide an example of a measure μ for which the Green potential of μ has non-tangential limit $+\infty$ almost everywhere on $\mathbb{S}$.

Finally, in Chapter 10 we introduce and investigate basic properties of weighted Bergman- and Dirichlet-type spaces of $\mathcal{H}$ -harmonic functions on $\mathbb{B}$, denoted respectively by $\mathcal{B}_\nu^p$ and $\mathcal{D}_\nu^p$. These spaces consist of the set of $\mathcal{H}$ -harmonic functions f on $\mathbb{B}$ for which f , respectively $|\nabla^h f|$, are in

$L^p((1 - |x|^2)^\gamma d\tau(x))$, $0 < p < \infty$, $\gamma > 0$, where τ is the invariant measure on $\mathbb{B}$ and ∇^h is the invariant gradient on $\mathbb{B}$. One of the main results of this chapter is that if $\gamma > (n - 1)$, then $f \in \mathcal{B}_\gamma^p$ if and only if $f \in \mathcal{D}_\gamma^p$ for all p , $0 < p < \infty$. In Section 10.4 we investigate the integrability of functions in $\mathcal{B}_\gamma^p$ and $\mathcal{D}_\gamma^p$ for $\gamma \leq (n - 1)$. This chapter also contains a discussion of Möbius invariant spaces of $\mathcal{H}$ -harmonic functions and the Berezin transform on $\mathbb{B}$. We conclude the chapter with three theorems of Hardy and Littlewood for $\mathcal{H}$ -harmonic functions, and the Littlewood–Paley inequalities for $\mathcal{H}$ -subharmonic functions.

At the end of each chapter, I have included a set of exercises dealing with the topics discussed. Many of these problems involve routine computations and inequalities not included in the text. They also provide examples relevant to the topics of the chapter. Also included are problems whose solutions may be suitable for possible publication. The latter are marked with an asterisk.

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Möbius Transformations

In this chapter we provide a brief review of Möbius transformations on n -dimensional Euclidean space $\mathbb{R}^n$ ($n \geq 2$). A good reference for these topics is the monograph by A. F. Beardon [11]. First, however, we begin with a review of notation that will be used throughout these notes.

1.1 Notation

For $x, y \in \mathbb{R}^n$ we let $\langle x, y \rangle = \sum_{j=1}^n x_j y_j$ denote the usual inner product on $\mathbb{R}^n$ and $|x| = \sqrt{\langle x, x \rangle}$ the length of the vector x . For $a \in \mathbb{R}^n$ and $r > 0$, the ball $B(a, r)$ and sphere $S(a, r)$ are given respectively by

$$B(a, r) = \{x \in \mathbb{R}^n : |x - a| < r\},$$

$$S(a, r) = \{x \in \mathbb{R}^n : |x - a| = r\}.$$

The unit ball and unit sphere with center at the origin will simply be denoted by $\mathbb{B}$ and $\mathbb{S}$ respectively.¹ The **one-point compactification** of $\mathbb{R}^n$, denoted $\hat{\mathbb{R}}^n$, is obtained by appending the point ∞ to $\mathbb{R}^n$. A subset U of $\hat{\mathbb{R}}^n = \mathbb{R}^n \cup \{\infty\}$ is open if it is an open subset of $\mathbb{R}^n$, or if U is the complement in $\hat{\mathbb{R}}^n$ of a compact subset C of $\mathbb{R}^n$. With this topology $\hat{\mathbb{R}}^n$ is compact.

For a subset D of $\mathbb{R}^n$, $\overline{D}$ denotes the closure of D , $\text{Int}(D)$ the interior of D , ∂D the boundary of D , and $\tilde{D}$ the complement of D in $\mathbb{R}^n$. Also if E and F are sets, $E \setminus F$ denotes the complement of F in E , that is, $E \setminus F = E \cap \tilde{F}$.

The study of functions of n -variables is simplified with the use of multi-index notation. For an ordered n -tuple $\alpha = (\alpha_1, \dots, \alpha_n)$, where each α_j is a non-negative integer, the following notational conventions will be used throughout:

¹ If we wish to emphasize the dimension n , we will use the notation $\mathbb{B}_n$ and $\mathbb{S}_n$ to denote the unit ball and sphere in $\mathbb{R}^n$.

$$|\alpha| = \alpha_1 + \cdots + \alpha_n, \quad \alpha! = \alpha_1! \cdots \alpha_n!, \quad x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n},$$

and

$$D^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$

If Ω is an open subset of $\mathbb{R}^n$, we denote by $C^k(\Omega)$, $k = 0, 1, 2, \dots$ the set of real-valued (or complex-valued) functions f on Ω for which $D^\alpha f$ exists and is continuous for all multi-indices α with $|\alpha| \leq k$. Thus $C^0(\Omega)$, or simply $C(\Omega)$, denotes the set of real-valued (or complex-valued) continuous functions on Ω , and $C^\infty(\Omega)$ the set of infinitely differentiable functions on Ω . Also, the set of functions $f \in C^k(\Omega)$ for which $D^\alpha f$, $|\alpha| \leq k$, has a continuous extension to $\overline{\Omega}$ will be denoted by $C^k(\overline{\Omega})$. If $f : \Omega \mapsto \mathbb{R}$, then the **support** of f , denoted $\text{supp } f$, is defined as

$$\text{supp } f = \overline{\{x \in \Omega : f(x) \neq 0\}}.$$

The set of continuous functions on Ω with compact support will be denoted by $C_c(\Omega)$. The notations $C_c^k(\Omega)$ and $C_c^\infty(\Omega)$ have the obvious meanings.

A linear transformation $A : \mathbb{R}^n \mapsto \mathbb{R}^n$ is said to be **orthogonal** if $|Ax| = |x|$ for all $x \in \mathbb{R}^n$. The set of orthogonal transformations of $\mathbb{R}^n$ will be denoted by $O(n)$. If A is represented by the $n \times n$ matrix $(a_{i,j})$, then A is orthogonal if and only if

$$\sum_{k=1}^n a_{i,k} a_{j,k} = \delta_{i,j} = \begin{cases} 1 & i = j, \\ 0, & i \neq j. \end{cases}$$

If $\psi(x) = (\psi_1(x), \dots, \psi_n(x))$ is a C^1 mapping of an open subset Ω of $\mathbb{R}^n$ into $\mathbb{R}^n$, then the derivative $\psi'(x)$ is the $n \times n$ matrix given by

$$\psi'(x) = \left(\frac{\partial \psi_i}{\partial x_j} \right)_{i,j=1}^n,$$

and the **Jacobian** J_ψ of the transformation ψ is given by $J_\psi(x) = \det \psi'(x)$.

1.2 Inversion in Spheres and Planes

Definition 1.2.1 The *inversion*² (or reflection) in the sphere $S(a, r)$ is the function $\phi(x)$ defined by

² Although we will mainly be interested in the case $n \geq 2$, the formulas for inversions in spheres and planes are still meaningful when $n = 1$.

$$\phi(x) = a + \left(\frac{r}{|x - a|} \right)^2 (x - a). \quad (1.2.1)$$

The inversion in the unit sphere $\mathbb{S}$ is the mapping $\phi(x) = x^*$ where

$$x^* = \begin{cases} \frac{x}{|x|^2} & x \neq 0, \infty, \\ 0 & x = \infty, \\ \infty & x = 0. \end{cases}$$

Thus (1.2.1) can now be rewritten as

$$\phi(x) = a + r^2(x - a)^*.$$

The reflection $\phi(x)$ is not defined at $x = a$. Since $|\phi(x)| \rightarrow \infty$ as $x \rightarrow a$ we set $\phi(a) = \infty$. Also, since $\lim_{|x| \rightarrow \infty} |\phi(x) - a| = 0$, we set $\phi(\infty) = a$. Thus ϕ is defined on all of $\hat{\mathbb{R}}^n$, and it is easily shown that ϕ is continuous in the topology of $\hat{\mathbb{R}}^n$. A straightforward computation also shows that $\phi(\phi(x)) = x$ for all $x \in \hat{\mathbb{R}}^n$. Thus ϕ is a one-to-one continuous map of $\hat{\mathbb{R}}^n$ onto $\hat{\mathbb{R}}^n$ satisfying $\phi(x) = x$ if and only if $x \in S(a, r)$.

In addition to reflection in a sphere we also have reflection in a plane. For $a \in \mathbb{R}^n$, $a \neq 0$, and $t \in \mathbb{R}$, the plane $P(a, t)$ is defined by

$$P(a, t) = \{x \in \mathbb{R}^n : \langle x, a \rangle = t\}.$$

By convention ∞ belongs to every plane $P(a, t)$.

Definition 1.2.2 The **inversion** (or **reflection**) in the plane $P(a, t)$ is the function $\psi(x)$ defined by

$$\psi(x) = x + \lambda a,$$

where $\lambda \in \mathbb{R}$ is chosen so that $\frac{1}{2}(x + \psi(x)) \in P(a, t)$.

Solving for λ gives

$$\psi(x) = x - 2[\langle x, a \rangle - t]a^*, \quad x \in \mathbb{R}^n. \quad (1.2.2)$$

For the mapping ψ we have

$$|\psi(x)|^2 = |x|^2 + O(|x|),$$

and as a consequence $\lim_{|x| \rightarrow \infty} |\psi(x)| = \infty$. Thus as above we define $\psi(\infty) = \infty$. With this definition the mapping ψ again satisfies $\psi(\psi(x)) = x$ for all $x \in \hat{\mathbb{R}}^n$. Thus ψ is a one-to-one continuous map of $\hat{\mathbb{R}}^n$ onto itself with $\psi(x) = x$ if and only if $x \in P(a, t)$. It is well known that each inversion (in a sphere or a plane) is orientation-reversing and conformal (see [11, Theorem 3.1.6]).

1.3 Möbius Transformations

Definition 1.3.1 A *Möbius transformation* of $\hat{\mathbb{R}}^n$ is a finite composition of inversions in spheres or planes.

Clearly the composition of two Möbius transformations is again a Möbius transformation, as is the inverse of a Möbius transformation. The group of Möbius transformations on $\hat{\mathbb{R}}^n$ is called the **general Möbius group** and is denoted by $\text{GM}(\hat{\mathbb{R}}^n)$. Although not immediately obvious, both translation and magnification by a constant are Möbius transformations. The translation $x \mapsto x + a$, $a \in \mathbb{R}^n$, is the composition of inversion in the plane $\langle x, a \rangle = 0$ followed by inversion in the plane $\langle x, a \rangle = \frac{1}{2}|a|^2$. Likewise, the magnification or scalar multiplication $x \mapsto kx$, $k > 0$, is also a Möbius transformation in that it is the inversion in $\mathbb{S}$ followed by the inversion in $S(0, \sqrt{k})$. Furthermore, every Euclidean isometry of $\mathbb{R}^n$ is a composition of at most $n + 1$ reflections in planes ([11, Theorem 3.1.3]).

We conclude this section by showing that every Möbius transformation maps a sphere or plane onto a sphere or plane. We will use the term “sphere” to denote either a sphere of the form $S(a, r)$ or a plane $P(a, t)$. Since every inversion ψ in a plane $P(a, t)$ can be written as

$$\psi(x) = x + \lambda a,$$

the mapping ψ clearly maps a “sphere” onto a “sphere.” To show that an inversion ϕ in a sphere $S(a, r)$ preserves “spheres,” it suffices to show that the mapping x^* preserves “spheres.”

For any set $E \subset \mathbb{R}^n$, we let $E^* = \{x^* : x \in E\}$. A set $E \subset \mathbb{R}^n$ is a sphere or a plane if and only if

$$E = \{x \in \mathbb{R}^n : b|x|^2 - 2\langle x, a \rangle + c = 0\},$$

where b and c are real and $a \in \mathbb{R}^n$. By convention, ∞ satisfies this equation if and only if $b = 0$, that is, E is a plane. Now it is easily seen that E^* has the same form with the roles of b and c reversed. Finally, it is an easy exercise to show that for $a \in \mathbb{R}^n$ and $r > 0$,

$$S^*(a, r) = \begin{cases} S\left(\frac{a}{(|a|^2 - r^2)}, \frac{r}{||a|^2 - r^2|}\right) & \text{if } 0 \notin S(a, r), \\ P(a, \frac{1}{2}) & \text{if } 0 \in S(a, r). \end{cases} \quad (1.3.1)$$

We conclude this section with one more useful formula that will be required later. If ϕ is inversion in the sphere $S(a, r)$, then a straightforward computation gives

$$|\phi(y) - \phi(x)| = \frac{r^2|y - x|}{|x - a||y - a|}. \quad (1.3.2)$$

For details on the above the reader is referred to [11].

Möbius Self-Maps of the Unit Ball

In this chapter we will provide a characterization of the Möbius transformations of $\hat{\mathbb{R}}^n$ mapping the unit ball $\mathbb{B}$ onto $\mathbb{B}$ that is similar to the characterization of the Möbius mappings of the unit disc in $\mathbb{C}$ onto itself.

In the complex plane $\mathbb{C}$, every analytic Möbius transformation ψ mapping the unit disc $\mathbb{D}$ onto itself can be written as $\psi(z) = e^{i\theta} \varphi_w(z)$, where for $w \in \mathbb{D}$,

$$\varphi_w(z) = \frac{w - z}{1 - \bar{w}z}.$$

The mappings $\varphi_w(z)$ satisfy $\varphi_w(0) = w$, $\varphi_w(w) = 0$, and $\varphi_w(\varphi_w(z)) = z$ for all $z \in \mathbb{D}$. Furthermore, the mapping $\varphi_w(z)$ also satisfies

$$1 - |\varphi_w(z)|^2 = \frac{(1 - |z|^2)(1 - |w|^2)}{|1 - \bar{w}z|^2}.$$

2.1 Möbius Transformations of $\mathbb{B}$

In this section we define an analogous family of Möbius transformations $\{\varphi_a : a \in \mathbb{B}\}$ mapping $\mathbb{B}$ onto $\mathbb{B}$ having the property that every Möbius transformation ψ mapping $\mathbb{B}$ onto itself can be written as $\psi = A \circ \varphi_a$, where $a \in \mathbb{B}$ and $A \in O(n)$. For $a \in \mathbb{B}$, we first set

$$\psi_a(x) = a + (1 - |a|^2)(a - x)^*. \quad (2.1.1)$$

Since the mapping ψ_a is a composition of Möbius transformations, ψ_a is a Möbius transformation of $\hat{\mathbb{R}}^n$ mapping 0 to a^* and a to ∞ . By a straightforward computation we have

$$|\psi_a(x)|^2 = \frac{|a - x|^2 + (1 - |a|^2)(1 - |x|^2)}{|a - x|^2}, \quad (2.1.2)$$

and as a consequence

$$|\psi_a(x)|^2 - 1 = \frac{(1 - |a|^2)(1 - |x|^2)}{|x - a|^2}. \quad (2.1.3)$$

From the above it follows immediately that ψ_a maps $\mathbb{B}$ onto $\hat{\mathbb{R}}^n \setminus \bar{\mathbb{B}}$.

We now define the mapping φ_a by

$$\varphi_a(x) = \psi_a(x)^* = \frac{\psi_a(x)}{|\psi_a(x)|^2}. \quad (2.1.4)$$

If we set¹

$$\rho(x, a) = |x - a|^2 + (1 - |a|^2)(1 - |x|^2) = |a|^2|a^* - x|^2, \quad (2.1.5)$$

then the mapping φ_a can be expressed as

$$\varphi_a(x) = \frac{a|x - a|^2 + (1 - |a|^2)(a - x)}{\rho(x, a)}. \quad (2.1.6)$$

As a consequence of (2.1.3)

$$1 - |\varphi_a(x)|^2 = \frac{(1 - |x|^2)(1 - |a|^2)}{\rho(x, a)}. \quad (2.1.7)$$

Thus φ_a is a Möbius transformation mapping $\mathbb{B}$ onto $\mathbb{B}$ with $\varphi_a(0) = a$ and $\varphi_a(a) = 0$. That φ_a maps $\mathbb{B}$ onto $\mathbb{B}$ follows immediately from the fact that ψ_a maps $\mathbb{B}$ onto $\hat{\mathbb{R}}^n \setminus \bar{\mathbb{B}}$ and that x^* maps $\hat{\mathbb{R}}^n \setminus \bar{\mathbb{B}}$ onto $\mathbb{B}$. We will shortly prove that φ_a also satisfies $\varphi_a(\varphi_a(x)) = x$ for all $x \in \mathbb{B}$. In the unit disk $\mathbb{D}$, for $z, w \in \mathbb{D}$, $\rho(z, w) = |1 - \bar{w}z|^2$ and the mappings $\varphi_w(z)$ as defined by (2.1.6) are precisely the functions $(w - z)/(1 - \bar{w}z)$.

One of the advantages of the mappings φ_a is that the function $(a, x) \mapsto \varphi_a(x)$ is not only continuous on $\bar{\mathbb{B}} \times \bar{\mathbb{B}}$ but also differentiable in each of the variables. At this point we will include several computations involving derivatives of the mappings φ_a that will be required in the proof of [Theorem 2.1.2](#) and also later in the sequel. Let $y_j(x)$ denote the j th coordinate of $y(x) = \varphi_a(x)$. Then by straightforward computations we have

$$\frac{\partial y_j}{\partial x_i}(0) = -\delta_{i,j}(1 - |a|^2), \quad \frac{\partial y_j}{\partial x_i}(a) = \frac{-\delta_{i,j}}{(1 - |a|^2)}, \quad (2.1.8)$$

$$\frac{\partial^2 y_j}{\partial x_i^2}(0) = (1 - |a|^2)[2a_j - 4a_i\delta_{i,j}]. \quad (2.1.9)$$

Hence

$$\varphi'_a(0) = -(1 - |a|^2)I \quad \text{and} \quad \varphi'_a(a) = -(1 - |a|^2)^{-1}I,$$

where I is the $n \times n$ identity matrix.

¹ In [11] the function $\sqrt{\rho(x, a)}$ is denoted by $[x, a]$.